

Junhong Zhang

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Shenzhen University

Computer Science • PhD candidate

Shenzhen

Educational Background

2022.09 to Present	College of Computer Science and Software Engineering, Shenzhen University Doctor (Direct PhD) in Computer Science and Technology
2018.09 to 2022.06	College of Computer Science and Software Engineering, Shenzhen University Bachelor; GPA: 4.21/4.50 (Rank 1 in major) in Computer Science and Technology (dual-degree program in Math and Computing Science).

Publications

Research focus: optimization, large-scale learning, representation learning. Published 14 papers (5 first-author: 4 in CAS Zone 1, 2 in CCF-A venues, and the rest are collaborative). Selected publications:

2025.04	J. Zhang , Z. Lai, H. Kong, and J. Yang, “Learning The Optimal Discriminant SVM with Feature Extraction,” IEEE Trans. Pattern Anal. Mach. Intell. (T-PAMI), vol. 47, no. 4, pp. 2897-2911, Apr. 2025. (CCF-A) ▶ An adaptive joint learning model of features and classification with complete convergence analysis and efficient engineering implementation. ▶ Project: https://github.com/Apple-Zhang/ODSVM-paper
2025.05	J. Zhang and Z. Lai, “Joker: Joint Optimization Framework for Lightweight Kernel Machines,” in Proc. 42nd Int. Conf. Mach. Learn. (ICML), 2025. (CCF-A) ▶ A general nonlinear learning framework for large-scale data: processes 10M data points with a single consumer-grade GPU of 10GB memory. ▶ Project: https://github.com/Apple-Zhang/Joker-paper
2026.02	J. Zhang , Z. Lai, J. Zhou, G. Liang, “NPSVC++: A Representation Learning Framework for Nonparallel Classifiers,” in IEEE Trans. Neural Networks Learn. Syst. (T-NNLS), 2026. (CAS Zone 1) ▶ A Pareto-optimal representation learning framework for nonparallel classifiers, improving classification accuracy by 1.5% over conventional cross-entropy. ▶ Project: https://github.com/Apple-Zhang/NPSVCP
2025.11	J. Zhang and Z. Lai, “LighTwSVM: Efficient Linear Nonparallel Classifier for Millions of Data,” Pattern Recognition (PR), Art. no. 112700, 2025. (CAS Zone 1) ▶ An efficient drop-in replacement for <code>sklearn.svm.LinearSVC</code>, about 2x faster than LIBLINEAR with roughly 50% lower memory usage. ▶ Project: https://github.com/Apple-Zhang/LighTwSVM-paper
2023.04	J. Zhang , Z. Lai, H. Kong, and L. Shen, “Robust Twin Bounded Support Vector Classifier with Manifold Regularization,” IEEE Trans. Cybern. (T-CYB), vol. 53, no. 8, pp. 5135-5150, 2023. (CAS Zone 1) ▶ A nonparallel classifier integrating a nonconvex robust metric and manifold structure to suppress noise, with global convergence guarantee. ▶ Project: https://github.com/Apple-Zhang/RMTBSVM-paper
2026.04	F. Chen, J. Zhang , Z. Lai, and H. Kong, “TaylorNet: Rethinking Monomial-based Graph Neural Networks with Taylor Expansion,” Pattern Recognition (PR), vol. 172, Art. no. 112748, 2026. (CAS Zone 1) ▶ Contributed generalization analysis: derived the generalization bound of GNNs formed by monomial graph filters with Taylor expansion and asymptotic analysis.

2026.06	F. Chen, J. Zhang , G. Liang, and Z. Lai, "A Design Principle for Graph Neural Networks," Pattern Recognition (PR), vol. 174, Art. no. 112999, 2026. (CAS Zone 1) Contributed the generalization theory: generalization bounds for graph-filter design of a family of GNNs via Rademacher complexity in a continuous-graph view.
2023.10	Z. Lai, X. Chen, J. Zhang , H. Kong, and J. Wen, "Maximal Margin Support Vector Machine for Feature Representation and Classification," IEEE Trans. Cybern. (T-CYB), vol. 53, no. 10, pp. 6700-6713, 2023. (CAS Zone 1) Verified the optimization details: derivations and convergence of the joint feature-extraction and sparse-selection optimization.
2026.07	F. Chen, Z. Lai, H. Lu, J. Zhang , Q. Wu, X. Luo, and H. Kong, "NSMNet: Stabilizing Linear State-space Memory for Breast Ultrasound Video Segmentation," ACM Multimedia (ACM-MM), 2026, Poster (accepted, to be published). Participated in idea formulation: identified rank vanishing issue of linear attention, designed stabilizing state-space memory for breast ultrasound video segmentation.
2026.07	Q. Wu, Z. Lai, X. Luo, H. Kong, F. Chen, J. Zhang , and G. Liang, "A Large-Scale Multi-Modal Benchmark and Robust Text-Free Inference for Breast Ultrasound Video Segmentation," ACM Multimedia (ACM-MM), 2026, Poster (accepted, to be published). Led problem formulation: annotation challenges for multi-modal breast ultrasound video segmentation, formulated a semi-supervised scheme via text-free inference.

Research Projects

2020-2021	Basic Experimental Project of the Student Innovation and Development Fund of Shenzhen University: Robust Twin Bounded Support Vector Classifier with Manifold Regularization (Sole Contributor) - As the sole contributor, independently completed the full loop from proposal, algorithm design, implementation, experimental validation and project conclusion. - Produced 1 IEEE T-CYB paper with open-sourced code (see publications); funded 10,000 CNY; successfully concluded.
2024-2025	Graduate Independent Innovation Achievement Cultivation Project of Shenzhen University: Efficient Big Data Algorithms Based on Kernel Methods (First Contributor) - As the first contributor, independently completed the full loop from proposal, algorithm design, implementation, experimental validation and project conclusion. - Produced 2 papers (ICML 2025, Pattern Recognition 2025) with open-sourced code (see publications); funded 50,000 CNY; successfully concluded.

Teaching Experience

2026 Spring	Multimedia Technology Fundamentals (top-up bachelor's course), Independent Lecturer •School of Continuing Education, Shenzhen University Independently taught the 12-week course (design, slides, lecturing), covering data compression principles and MPEG/H.264/HEVC video coding. Designed 2 hands-on labs and 3 comprehensive tests, forming a complete "lecturing–assessment" closed loop.
2020-2025	SVM Special Topic Lecturer (Machine Learning course) •Shenzhen University Delivered the SVM (Support Vector Machine) special topic in the Machine Learning course each year, with lecture notes and recorded videos (SVM-Intro).
2021-2023	Machine Learning Course Teaching Assistant (undergraduate) •Shenzhen University

Skills

Programming | Python, C/C++, MATLAB; PyTorch, NumPy/SciPy, Scikit-learn; Git, Linux

Languages | Chinese (native); English: academic writing and communication

Academic Service

- › Conference reviewing: ICME'24, AAAI'25, ICCV'25, CVPR'26, ICML'26, ACM-MM'26 (11 reviews in total)
- › Journal reviewing: reviewer for TKDE, T-NNLS, T-PAMI (21 reviews completed)

Awards and Honors

University-level Awards / Social Scholarships

- › 2021 Pengcheng Scholarship, Shenzhen University
- › 2021, 2024 Scholarship of the Excellent PhD Candidate Selection and Funding Program, College of Computer Science and Software Engineering, Shenzhen University
- › 2022 Excellent Undergraduate Graduate and Honorary Bachelor, Shenzhen University
- › 2023 Scholarship of the Top-Talent Cultivation Program, Shenzhen University

Academic Competition Awards:

- › 2020 Honorable Mention (Second Prize) in the Mathematical Contest in Modeling (MCM)
- › 2020 Meritorious Winner (First Prize) in the Mathematical Modeling International Contest (China)
- › 2021 Finalist (Outstanding Winner nomination, top 2% worldwide) in the Mathematical Contest in Modeling (MCM)